Abhishek Kumar Singh

Embedded Systems | Edge AI | IoT
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PROFILE

B.Tech ECE student specialising in IoT, with hands-on projects across ESP32 firmware, signed sensor telemetry, Python model tooling, and C++ inference. Interested in embedded systems and Edge AI internship opportunities.

EDUCATION

Madan Mohan Malaviya University of Technology, Gorakhpur
B.Tech - Electronics & Communication Engineering (IoT specialisation)
2025-2029 | In progress

SELECTED PROJECTS

AeroNode | C++17 adaptive inference runtime

Implementation covers multi-exit PyTorch training, separate ONNX stage exports, persistent C++ sessions, explicit tensor buffers, entropy-based early exits, and ZeroMQ runtime-state handling. Downstream stage skipping and numerical parity are verified.

Experimental MVP complete through Phase 8. Frozen policy evaluation: mean 2.1052 ms versus 2.8139 ms monolithic; accuracy 72.6857% versus 72.9400% on 1,966 images separate from threshold fitting. Scope: Windows CPU, batch 1, one thread, three repetitions. All 127 unit/integration cases pass. Local HTTP/PostgreSQL telemetry persisted and replayed 35,388 records without duplicates.

JOCKY | Defensive digital-forensics platform

Rust language/compiler and deterministic IR, bounded collectors, Sigma/YARA-X analysis, correlation/timelines, signed Ed25519 evidence checkpoints, and a React investigation console. Repository records 18 Rust, 29 Python and 7 frontend unit tests, plus a Playwright investigation journey. Full console runs locally; public site provides a synthetic lab preview.

TrackSure | Signed IoT shipment telemetry

Working prototype connects a battery-powered ESP32-S3, DHT11/MPU6050/LDR sensors, and ATECC508A event signing to a verifying gateway, durable SQLite outbox, FastAPI API, PostgreSQL storage, and admin/user dashboards.

Repository records real hardware-to-admin delivery and a retained 30-record event/hash ledger. Manual gateway startup, endurance testing, and independent backend signature auditing remain further work.

TECHNICAL SKILLS

Languages: C, C++17, Python, Rust, TypeScript
Embedded: ESP32-S3, I2C sensors, MPU6050, ATECC508A
Model/runtime: PyTorch, ONNX, ONNX Runtime, ZeroMQ, CMake
Systems: FastAPI, PostgreSQL, SQLite, React, Sigma, YARA-X, Git/GitHub